Smothering with Cervical Fracture Simulating Natural Death: A Case Report

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Abstract:

Deaths due to accidental falls in intoxicated individuals are common in forensic practice and may conceal homicidal acts. A meticulous medico legal autopsy helps to distinguish homicide from suicide. This case report deals with a 37-year-old male who was brought for medico-legal autopsy following an alleged accidental fall while under the influence of alcohol. This case underscores the indispensable role of thorough forensic autopsy in detecting concealed homicides and highlights the importance of maintaining a high index of suspicion in deaths initially reported as accidental, particularly in the presence of intoxication and ambiguous circumstances.

Keywords: Autopsy, Asphyxia, Homicide, Forensic investigation, Smothering

Introduction:

Deaths reported as accidental or natural amount to a major proportion of medico-legal cases in India, particularly those involving alcohol intoxication, alleged falls, or sudden deaths occurring at home. While many such deaths are genuinely accidental, a significant number represent concealed homicides, where intentional violence is hidden by misleading circumstances or narratives. [1] In India, socio-legal factors such as familial influence, social stigma, delayed reporting, lack of eyewitnesses, and initial acceptance of circumstantial histories often attribute to the misclassification of homicidal deaths as accidental or natural. [2] Although exact national data on concealed homicide are unavailable, discrepancies between hospital death certification and post-mortem findings, as well as revisions of manner of death following autopsy, suggest that homicidal deaths may be unreported in official crime statistics, including those published by the National Crime Records Bureau (NCRB). [3]

Indian law mandates medico-legal investigation of suspicious or unnatural deaths under Section 194 of Bharatiya Nagarik Suraksha Sanhita (BNSS) 2023, which authorizes the police to conduct inquests and forward the body for post-mortem examination under suspicious circumstances. [4] However, the inquest report is primarily descriptive and does not determine the cause of death, making the forensic autopsy the cornerstone of medicolegal death investigation. Judicial precedents have consistently emphasized that medical evidence derived from autopsy findings carries substantial probative value in criminal trials, particularly in differentiating homicide from suicide or accident.

Smothering, a form of mechanical asphyxia, is especially challenging to diagnose due to the frequent absence of obvious external injuries. The diagnostic difficulty is further complicated when intoxication is involved, as alcohol consumption may plausibly explain collapse, loss of consciousness, or accidental trauma. Associated injuries such as cervical spine fractures may be overlooked clinically or misattributed to accidental falls unless specifically sought during post-mortem examination. [5]

A meticulous forensic autopsy, involving systematic external examination, layered neck dissection, evaluation of cervical vertebrae, and correlation with circumstantial evidence, is

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essential to identify subtle indicators of violence. Autopsy findings often serve as the decisive factor in classifying the manner of death and initiating further criminal investigation under relevant sections of Bharatiya Nagarik Suraksha Sanhita (BNSS). [6] The present case illustrates the pivotal role of forensic autopsy in disclosing a concealed homicide initially presented as an accidental death, reinforcing its indispensable function in ensuring accurate cause and manner of death determination and upholding the interests of justice.

Case presentation:

A 37-year-old male was brought for medico-legal autopsy following death under allegedly accidental circumstances. According to police records and family statements, the deceased had attended a party with friends on the previous night, during which he reportedly consumed excessive alcohol and fell on the road. His friends claimed to have escorted him home and made him sleep in his room around 1:00 AM. In the early hours of the following morning, he was found unresponsive by his wife and was immediately taken to the hospital, where he was examined by the duty medical officer and declared brought dead. Consequently, a medico-legal case was registered, and the body was sent for post-mortem examination. There was no documented history of chronic illness, prior trauma, or suicidal ideation. The circumstances were initially presumed to indicate an accidental death related to alcohol intoxication.

Case findings:

External findings

External examination revealed multiple contusions over the face involving on the nose, cheek, lips and neck(Fig 1). Nail scratch marks were noted over the neck region (Fig 2), suggesting a struggle. No injuries consistent with a fatal accidental fall were identified.

Internal findings

On internal examination of the neck, diffuse contusions of the neck muscles were observed, indicative of forceful compression (Fig 3). A fracture of the right side of the hyoid bone was identified (Fig 4). Further examination revealed a dislocation fracture of the cervical spine at the C2–C3 level, accompanied by disruption of the spinal cord. (Fig 5) These findings were incompatible with the alleged history of a simple fall due to intoxication and strongly suggested external violence. After correlating the autopsy findings with the circumstantial evidence, the cause of death was certified as asphyxia due to smothering with associated cervical fracture.

Additionally, the viscera of the deceased was subjected to chemical analysis in Telangana Forensic Science Laboratory, Hyderabad which was negative for poisonous substances as well as alcohol.

Discussion:

The present case highlights the complex interplay between circumstantial history and forensic evidence in the determination of cause and manner of death. Deaths that are initially reported as accidental particularly in the setting of alcohol intoxication may conceal homicidal violence unless scrutinized through detailed medico-legal investigation. [7] In many forensic autopsy series from India, violent asphyxial deaths comprise a significant proportion of unnatural fatalities, highlighting the need for vigilance when evaluating such cases. Retrospective analyses from forensic departments in India have reported that asphyxial deaths account for approximately 7–8% of autopsied medicolegal deaths, with hanging, drowning, and other asphyxial mechanisms featuring prominently. [8]

Although smothering and other non-strangulation asphyxial methods are less common, they are predominantly homicidal in nature. A tertiary-centre study by Kumar Guntheti (2021) observed that all cases of smothering, throttling, and strangulation in their cohort were homicidal, whereas drowning and hanging showed mixed manners.[9] Studies have demonstrated that an appreciable fraction of asphyxia homicides may be first identified at autopsy when clinical or death

scene histories suggest alternative mechanisms, reinforcing the universal significance of post-mortem evaluation. [10]

Homicidal fatalities in India have been reported to comprise approximately 5–6% of all medico-legal autopsies. This may understate the actual burden due to misclassification prior to autopsy.[11] Additionally, Nasim Irshad (2022) analysed the forensic autopsy data from tertiary care centers and he stated that neck and head injuries including cervical spine trauma are frequently encountered in homicidal victims and are critical markers for violent death. [12] In this context, the identification of a cervical vertebral fracture alongside soft-tissue signs of asphyxia made it possible to challenge the initial accidental death narrative and accurately classify the death as homicidal.

A thorough post-mortem examination is valuable not only for determining the exact cause of death but also for confirming the manner of death, which is a crucial in criminal investigation and decision-making. Autopsy allows for systematic documentation of injuries, comprehensive internal examination, collection of forensic samples, and correlation with investigative findings, thereby preventing misclassification of homicides as accidents or natural deaths. In India, where medico-legal autopsy findings frequently form the primary evidence for legal proceedings, the role of a forensic pathologist in differentiating homicide from accident or suicide is indispensable.

Conclusion:

Forensic autopsy remains indispensable in accurately determining the cause and manner of death, distinguishing homicide from suicide or accident, and preventing misclassification of unnatural deaths. Such thorough medicolegal evaluation not only aids in the administration of justice but also reinforces the responsibility of forensic practitioners to maintain a high index of suspicion in equivocal deaths, ensuring that concealed homicides do not go undetected.

Reference:

1. Prahlow SP, Cohle S, Shattuck B, Prahlow JA. Homicides Disguised as Staged Suicides. Acad Forensic Pathol. 2020;10(2):104-112.
2. Rupesh KA, Sudha R, Miriyala R, S NV. Breaking the silence: Exploring the emotional and social hazards of forensic pathology practice in India. Indian J Forensic Community Med. 2024;11(1):35-37.
3. Kurz SD, Sido V, Herbst H, Ulm B, Salkic E, Ruschinski TM, Buschmann CT, Tsokos M. Discrepancies between clinical diagnosis and hospital autopsy: A comparative retrospective analysis of 1,112 cases. PLoS One. 2021;16(8):e0255490.
4. Government of India. Bharatiya Nagarik Suraksha Sanhita, 2023. New Delhi: Ministry of Law and Justice.
5. Kusumaningrat, Donald & Yapiter, Jesslyn. Forensic Examination in a Homicide Case Due to Suffocation: A Case Report. Med. 2025;14:213-221.
6. Zain Elabedin, S., Hassan Gaballah, M., Habiballah, M. et al. Sudden death due to silent cerebral arteriovenous malformations: case report. Egypt J Forensic Sci 2025;15:53
7. Wang H, Xu H, Li W, Li B, Shi Q, Ma K, Xiao B, Chen L. Forensic appraisal of death due to acute alcohol poisoning: three case reports and a literature review. Forensic Sci Res. 2019;5(4):341-347.
8. Iqbal, Farzand & Ullah, Faqeer & Ahmed, Iftikhar & Abdul Samad, Abrar & Tanoli, Aftab & Aziz, Ijaz. An Autopsy-Based Retrospective Analysis of Asphyxia-Related Deaths. BMC Med. Sci. 2024;5:92-95.
9. Kumar Guntheti , B. Pattern of asphyxial deaths: A medicolegal study. J. Indian Acad. Forensic Med. or J Indian Acad Forensic Med. 2021;43(1):55-59.
10. Begum, Asma & Khan, Nashid & Shafiuzzaman, AKM & Shahid, Farhana & Anam, AM & Ahmed, Khan & Begum, Rowson & Fahmi, Soniya. Suicidal Death due to Hanging. Delta Med. Coll. J. 2017;5:89.

11. Rathod VV, Choudhary UK, Ghormade PS, Keoliya AN. Study of socio-demographic profile of victims in cases of deaths due to homicide. Indian J Forensic Community Med. 2020;7(2):66-71.

12. Nasim Irshad, Naseem Akhter, Asiya Parveen, Aftab Alam Tanoli, Amna Shahid, & Mobin Inam Pal. Analysis of injury patterns in homicide victims: a forensic autopsy study. J Popul Ther Clin Pharmacol, 2022;29(04):4559-4564.

Figures

Figure 1 – Multiple Contusions over the face and lips

Figure 2 – Nail scratch marks on the side of the neck

Figure 3 – Contusion of neck muscles

Figure 4 – Fractured Hyoid bone

Figure 5 - Dislocation fracture of the cervical spine at the C2-C3 level with spinal cord disruption